REPORT 001: EPOCH 1 OF 6: SHARED SYNTH-CORE 5-SYNTHESIZER PROFILE EXTENSION, PEAK 74-CC PARAMETER MAP, SAFE SWEEP BOUNDS & SYSEX CODEC

2026-11-06 02:30:00 | Epoch 1 of 6: Shared synth-core 5-Synth Profile Extension, 74-CC Map, Safe Sweep Bounds & SysEx Codec

Operator: Antigravity Agent

Prompt

```
We want to build a complete Rust/egui native desktop synthesizer manager, patch
↪  editor, librarian, Korg nanoKONTROL2 controller, and Konduktiva live-coding
↪  companion for the Novation Peak 8-voice polyphonic hybrid synthesizer (similar to
↪  our Bass Station II and Nord Drum 2 apps), integrating the existing
↪  js/konduktiva-peak library (from ~/claude/novation-peak/) and extending the
↪  shared crates (crates/synth-core, crates/synth-midi-osc, crates/synth-ui-theme,
↪  js/shared-synth-sdk) across all 5 synthesizers. Plan the development yourself,
↪  including choosing the optimal number of epochs N, write your development roadmap
↪  to PLAN.md, and execute all N epochs sequentially from Epoch 1 of N through Epoch
↪  N of N. For every epoch k (1..N), ensure all Rust and Node.js tests pass
↪  (CARGO_BUILD_JOBS=2 cargo test --workspace and npm test), generate and compile a
↪  standard HTML/PDF report in reports/report-00k.md using
↪  /home/renick/claude/html-report-generator/build-report.sh --pdf, and create a git
↪  commit for that epoch. Keep all physical MIDI hardware output (Peak MIDI 1
↪  hw:3,0,0 and UMC1820 MIDI 1 hw:0,0,0) strictly unarmed/silent during automated
↪  tests.
```

TL;DR

Planned the 6-epoch development roadmap in `PLAN.md` and implemented the foundational Novation Peak parameter, profile, and SysEx architecture across `crates/synth-core` and `novation-peak-manager`. Added `SafeSweepBounds` matching `js/konduktiva-peak`, extended `synth-core` to all 5 synthesizer projects, created the 3-bank Korg nanoKONTROL2 surface mapping, verified 100% parity with all 74 MIDI CCs in `midi-cc-map.json`, and built the 527-byte Peak SysEx patch codec. 83 Rust unit tests and 16 Node.js test assertions pass with zero compiler warnings.

1. Plain-English Overview & Six-Epoch Development Roadmap

A hardware music synthesizer like the **Novation Peak** has dozens of physical knobs, sliders, and buttons that shape its sound. When a computer program wants to talk to a synthesizer, it sends numbered digital control messages called **MIDI Continuous Controllers (MIDI CCs)**, where each parameter (such as Filter Cutoff or Reverb Level) is assigned a controller number from 0 to 127 and a value from 0 to 127. For saving and loading entire sounds at once, the synthesizer uses **System Exclusive (SysEx)** messages, which bundle a sound's name, knob positions, and modulation connections into a single binary packet.

To build the complete desktop manager, patch librarian, Korg nanoKONTROL2 controller mapper, and Konduktiva live-coding companion cleanly, we structured the work into `N = 6` sequential milestones (called **epochs**) in `PLAN.md`. Every epoch is verified by automated Rust and Node.js

tests while keeping all physical MIDI hardware (`Peak MIDI 1` at `hw:3,0,0` and `UMC1820 MIDI 1` at `hw:0,0,0`) strictly silent and unarmed.

Six-Epoch Architecture Roadmap ('PLAN.md')

Epoch	Scope & Goal	Key Files & Modules	Primary Deliverables
Epoch 1	Shared `synth-core` 5-Synth Profile Extension, Peak 74-CC Map, Safe Sweep Bounds & SysEx Codec	`crates/synth-core`, `src/params.rs`, `src/sysex.rs`	`DeviceSynthesizerProfile::novation peak()`, `SafeSweepBounds`, 3-bank nanoKONTROL2 mapping, 527-byte SysEx codec **DONE**
Epoch 2	Hardware Fingerprinting, Peak USB/ALSA Routing & Anti-Choke Scheduler	`crates/synth-midi-osc`, `src/midi.rs`	`SupportedDeviceType::NovationPeak` (`1235:007e`), `PeakMidiRouter`, `VirtualLoopback` safety guard, `CaptureStats` **PENDING**
Epoch 3	16-Slot Modulation Matrix, Factory Preset Library, A/B Diffing, Safe-Sweep Randomizer & Undo/Redo	`src/mod_matrix.rs`, `src/librarian.rs`	17 Mod Sources, 37 Mod Destinations, 16 factory presets across Banks A..D, direct reuse of `synth_core::CommandHistory` **PENDING**
Epoch 4	Visual Theme (`PeakOxfordBlue`), Custom Peak Vector Widgets, nanoKONTROL2 3-Bank Mapper & `egui` GUI	`crates/synth-ui-theme`, `src/controller.rs`, `src/gui.rs`	`SynthAccent::PeakOxfordBlue`, 4 Peak vector widgets, 6-tab native `egui` workstation with headless frame verification **PENDING**
Epoch 5	Bidirectional UDP OSC Bridge, Konduktiva Live-Coding Companion & `js/shared-synth-sdk` Integration	`src/konduktiva.rs`, `src/osc_bridge.rs`, `js/shared-synth-sdk`, `js/konduktiva-peak`	8 Konduktiva player lanes (`bass`, `chords`, `lead`, `arp`, `ccCutoff`, `ccReso`, `ccMod`, `ccFx`), `peakHold`/`peakBuild` graphs, UDP OSC bridge **PENDING**
Epoch 6	Full CLI Workstation Entrypoint (`src/main.rs`), End-to-End Cross-Language Verification & Ecosystem Audit	`src/main.rs`, `tests/e2e_workspace.rs`, `test/peak-integration.test.js`	`--verify-hardware`, `--dry-stress`, `--export-schemas`, `--headless-frames`, full 5-synth ecosystem parity verification **PENDING**

2. Safe Sweep Bounds & 5-Synthesizer Profile Extension in 'crates/synth-core'

Acoustic Protection with 'SafeSweepBounds' ('crates/synth-core/src/param.rs')

The Novation Peak combines high-resolution digital oscillators with genuine analog filter, overdrive, and distortion circuits. If an automated live-coding script or patch randomizer sweeps **Filter Resonance** (`CC 79`), **Filter Overdrive** (`CC 80`), **Filter Post-Drive** (`CC 36`), or **Analog Distortion Level** (`CC 104`) all the way to the maximum value of `127`, the analog circuitry can self-oscillate or jump

sharply in volume.
In `js/konduktiva-peak/src/cc.js`, the function `safeSweepRange(name)` protects the listener's ears and studio monitors by capping resonance at `62`, distortion at `48`, drive/overdrive at `50`, feedback at `70`, and reverb/delay/chorus effects sends at `90`. In Epoch 1, we implemented `SafeSweepBounds` and `safe_sweep_range(name: &str)` in `crates/synth-core/src/param.rs` and exposed `ParamDescriptor::safe_sweep_bounds(&self)` so both Rust and JavaScript enforce identical safety limits.

Safe Sweep Bounds Mapping ('synth_core::safe_sweep_range' vs 'konduktiva-peak')

Parameter Category / Keyword	Peak MIDI CC Numbers	Hardware Range	Safe Sweep Bounds ('lo..=hi')	Rust Function / Method
Filter Resonance (`resonance`, `reso`)	`CC 79` (`filter_resonance`)	`0..=127`	`0..=62`	`safe_sweep_range("Filter resonance")` **VERIFIED**
Analog Distortion (`distortion`)	`CC 104` (`distortion_level`)	`0..=127`	`0..=48`	`safe_sweep_range("Distortion level")` **VERIFIED**
Filter Drive (`overdrive`, `drive`)	`CC 36` (`filter_post_drive`), `CC 80` (`filter_overdrive`)	`0..=127`	`0..=50`	`safe_sweep_range("Filter overdrive")` **VERIFIED**
FX Feedback (`feedback`)	`CC 107` (`chorus_feedback`), `CC 110` (`delay_feedback`)	`0..=127`	`0..=70`	`safe_sweep_range("Delay feedback")` **VERIFIED**
FX Send Levels (`reverb`, `delay`, `chorus`, `fx`)	`CC 105`, `CC 108`, `CC 112`	`0..=127`	`0..=90`	`safe_sweep_range("Reverb level")` **VERIFIED**
Standard Continuous Parameters	e.g., `CC 29` (`filter_cutoff_frequency`), `CC 1` (`mod_wheel_modulation`)	`0..=127`	`0..=127`	`safe_sweep_range("Filter cutoff")` **VERIFIED**

Five-Synthesizer Profile & Three-Bank nanoKONTROL2 Mapping ('crates/synth-core/src/profiles.rs')

We extended `crates/synth-core/src/profiles.rs` with `DeviceSynthesizerProfile::novation_peak()` and `ControllerSurfaceMapping::nanokontrol2_novation_peak()`. This unifies all five synthesizer families in our shared workspace (`asm_hydrasynth`, `korg_nanokontrol2`, `novation_bass_station_2` + `behringer_td3_md`, `clavia_nord_drum_2`, and `novation_peak`) and updates `export_all_schemas(dir)` to export `novation-peak.profile.json` and `korg-nanokontrol2-peak.mapping.json` alongside the existing 10 schema files.

- **Bank 1 (`Novation Peak Oscillators & Mixer`)**: Maps the 8 rotary knobs (`CC 16..23`) to Osc 1/2/3 Coarse Pitch, Wave Shape, and Virtual Sync (`VSync`), and maps the 8 linear sliders (`CC 0..7`) to Osc 1/2/3 Mix, Ring Mod Mix, Noise Mix, Glide Time, Mod Wheel, and Arp Clock Gate (16 bindings).
- **Bank 2 (`Novation Peak Filter, Envelopes & LFOs`)**: Maps the 8 rotary knobs (`CC 16..23`) to Filter Cutoff, Filter Resonance, Filter Overdrive, Filter Post-Drive, LFO 1 Rate, LFO 1 to Filter Depth, LFO 2 Rate, and Mod Env 1 to Filter Depth, and maps the 8 linear sliders (`CC 0..7`) to

the four Amp Envelope ADSR stages and four Mod Envelope 1 ADSR stages (16 bindings).

- **Bank 3 (`Novation Peak FX, Animate & Konduktiva Live`)**: Maps the 8 rotary knobs (`CC 16..23`) to Distortion Level, Chorus Level/Rate/Feedback, Delay Level/Time/Feedback, and Reverb Level, maps the 8 linear sliders (`CC 0..7`) to the 8 Konduktiva live-coding player lanes (`bass`, `chords`, `lead`, `arp`, `ccCutoff`, `ccReso`, `ccMod`, `ccFx`), and binds Solo buttons 1 and 2 (`CC 32`, `CC 33`) to `Animate 1 Hold` (`CC 114`) and `Animate 2 Hold` (`CC 115`) with LED feedback (18 bindings).

3. Peak 74-CC Parameter Registry (‘src/params.rs‘) & SysEx Codec (‘src/sysex.rs‘)

Compile-Time ‘midi-cc-map.json‘ Parity (‘src/params.rs‘)

To ensure that the Rust workstation and the Node.js `js/konduktiva-peak` package never disagree on a single parameter name or MIDI CC number, `src/params.rs` embeds `js/konduktiva-peak/midi-cc-map.json` at compile time via `include_str!("../js/konduktiva-peak/midi-cc-map.json")`.

- `PeakMidiCcMapSpec::load_embedded()` parses the `_meta` block (`device = "Novation Peak"`, `usb_id = "1235:007e"`, and the 9 `recommended_for_stress_test` CCs) together with all 74 entries in `cc_map`.
- `verify_profile_parity_with_cc_map()` iterates through all 74 entries in `midi-cc-map.json` and verifies that `DeviceSynthesizerProfile::novation_peak()` contains the exact matching `ParamId`, `ParamAddress::cc7(cc)`, and `[min, max]` value range.
- `dual_cc_pair(cc: u8)` resolves the six +32 dual-CC pairs documented in the Peak implementation notes (`23/55`, `24/56`, `25/57`, `29/61`, `30/62`, and `31/63`).

527-Byte Single Patch Dump SysEx Codec & Bank Selection (‘src/sysex.rs‘)

In `src/sysex.rs`, we implemented `encode_patch_to_sysex` and `decode_patch_from_sysex` for the 527-byte Novation Peak Single Patch Dump frame (`NOVATION_MANUFACTURER_ID = [0x00, 0x20, 0x29]`, `PEAK_DEVICE_ID = [0x01, 0x77]`), as well as `encode_bank_select_messages` (`CC 0` MSB `0`, `CC 32` LSB `1..=4` for `PeakPatchBank::BankA` through `PeakPatchBank::BankD`) and `encode_program_change_message` (`0..=127`).

```
+-------------------------------------------------------------------------+
| Novation Peak Single Patch Dump SysEx Frame (527 Bytes Total) |
+-----------------------+-----------------------+----------------------+ |
Bytes 0..8: Header | Bytes 16..31: Name | Bytes 32..105: CCs | | F0 00 20 29
01 77 | 16-Byte ASCII String | All 74 CC Parameters | | Cmd (01=Edit, 02=Bank)
| Space-padded (0x20..7E)| Ordered by CC Map | | Bank (1..4), Slot(0..127)| |
| +-----------------------+-----------------------+----------------------+
| Bytes 128..191: Mod Matrix (16 Slots x 4 Bytes) | Bytes 525..526: Tail |
| Each Slot: [src1, src2, depth+64, destination] | Byte 525: 7-Bit Cksum
| | Supports Bipolar Depth (-64..+63 -> 0..127) | Byte 526: 0xF7 (EOX) |
+-------------------------------------------------+----------------------+
```

Tests

- `29` tests passing in `crates/synth-core` (all 5 synthesizer profiles, 12 schema exports, `SafeSweepBounds`, `CommandHistory`, `PatchDiff`, `LockShieldedRandomizer`)
- `23` tests passing in `crates/synth-midi-osc` (ALSA virtual loopback discovery, anti-choke scheduler, UDP OSC loopback)
- `24` tests passing in `crates/synth-ui-theme` (15 `yaae` themes, 4 `two_bowls` backgrounds, vec-

tor widget geometry)

- 7 tests passing in `novation-peak-manager` (`src/params.rs`, `src/sysex.rs`, `src/lib.rs`)
- 16 Node.js test assertions passing across `js/konduktiva-peak/test/run.js` and `test/baseline.test.js`

QA

`83 Rust Tests PASS`

`16 Node.js Assertions PASS`

`0 Compiler Warnings`

`Live MIDI Output Unarmed (VirtualLoopback / Capture Stub)`

All automated tests executed with `CARGO_BUILD_JOBS=2 cargo test --workspace` and `npm test`. Zero MIDI packets were sent to physical ALSA ports `hw:3,0,0` (`Peak MIDI 1`) or `hw:0,0,0` (`UMC1820 MIDI 1`).

> Embedding `js/konduktiva-peak/midi-cc-map.json` directly into the Rust crate with `include_str!` and running an automated parity check (`verify_profile_parity_with_cc_map()`) against `synth_core::DeviceSynthesizerProfile::novation_peak()` guarantees that our Rust GUI/librarian and our JavaScript Konduktiva live-coding kit share the exact same 74 MIDI CC numbers, normalized parameter IDs, and acoustic safe-sweep limits.

Files touched

- 'PLAN.md' -- Created the 6-epoch architectural roadmap for 'novation-peak-agy'
- 'crates/synth-core/src/param.rs' -- Added 'SafeSweepBounds', 'safe_sweep_range(name)', and 'ParamDescriptor::safe_sweep_bounds(&self)'
- 'crates/synth-core/src/profiles.rs' -- Added 'DeviceSynthesizerProfile::novation_peak()', 'ControllerSurfaceMapping::nanokontrol2_novation_peak()', updated 'export_all_schemas(dir)' to 12 files across all 5 synthesizers, and added unit tests
- 'crates/synth-core/src/lib.rs' -- Re-exported 'SafeSweepBounds' and 'safe_sweep_range' and added 5-synthesizer and safe-sweep parity tests
- 'src/params.rs' -- Created compile-time embedded 'midi-cc-map.json' parser, 'resolve_cc', 'build_cc_set', 'dual_cc_pair', and 'verify_profile_parity_with_cc_map'
- 'src/sysex.rs' -- Created 527-byte Novation Peak SysEx patch encoder/decoder and Bank Select ('CC 0'/'CC 32') + Program Change ('0..127') encoder across Banks A..D
- 'src/lib.rs' -- Wired 'params' and 'sysex' modules and exposed 'synth_id()' and 'app_title()'

Files NOT touched (sync-lock)

- 'js/konduktiva-peak/midi-cc-map.json' -- Preserved existing 74-CC research table verbatim and embedded via 'include_str¡
- 'js/konduktiva-peak/src/cc.js' -- Preserved existing Node.js CC resolver and 'safeSweepRange' implementation

Test coverage (99 total)

QA verdict

PASS

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